

Baiyu Shi

(510) 219-0132 • baiyushi@stanford.edu • [Personal Webpage](#)

EDUCATION

Stanford University
3rd year PhD. Candidate In Mechanical Engineering

Stanford, CA
September 2023 - Present

- GPA: **4.123/4.3**.
- Advisor: Professor Zhenan Bao

University of California, Berkeley
B.S. Bioengineering, Electrical Engineering and Computer Science (High Honors)

Berkeley, CA
September 2019 - May 2023

- GPA: **3.965/4.00**.
- Advisors: Professor Robert J. Full, Professor Ken Goldberg, and Professor Gerard Marriott.
- Awards: UC Berkeley **Departmental Citation** in Bioengineering, 2023 Outstanding Graduate Student Instructor.

RESEARCH INTERESTS

My research interests lie at the intersection of high-performance, scalable electronic skin and tactile intelligence for robotics, with a focus on contact-aware dexterous manipulation and safe physical interaction.

SELECTED PUBLICATION

- [2] **DexSkin: High-Coverage Conformable Robotic Skin for Learning Contact-Rich Manipulation.** (Oral ~ 5.7%)
S. Wistreich †, **B. Shi** †, S. Tian †, S. Clarke, M. Nath, C. Xu, Z. Bao, J. Wu *CoRL 2025*
- [1] **Bagging by Learning to Singulate Layers Using Interactive Perception.** (Best Industrial Robotics Finalist)
L. Y. Chen, **B. Shi**, R. Lin, D. Seita, A. Ahmad, R. Cheng, T. Kollar, D. Held, K. Goldberg *IROS 2023*

RESEARCH EXPERIENCE

Robotic Tactile Sensing and Dexterous Manipulation (DigiSkin, DexSkin Projects)

Stanford University

PI: Prof. Zhenan Bao

Jan 2024- Present

- Developed the DigiSkin platform, including its manufacturing process and large-scale high-speed readout, for tailorable and stretchable electronic skins with **high sensitivity, wide dynamic range, and low hysteresis**, achieving leading **sub-millimeter spatial resolution** and **large-area tactile coverage**.
- Integrated DigiSkin with robotic grippers and dexterous hands to enable **conformal tactile sensing over curved surfaces**; applied tactile cues in **imitation learning and residual RL** policies for pen reorientation, box packaging, and berry sorting.
- Designed and fabricated a **wearable DigiSkin glove** for real-time full-hand tactile sensing, enabling accurate classification of 40+ everyday object with **strong cross-user generalization** (N=4).

Neuromechanics and Robot Learning (Lizard Neuromechanics, AutoBag, SLIP-Bagging, AVSI Projects) **UC Berkeley**

PI: Prof. Robert J. Full & Prof. Ken Goldberg

Jun 2021- May 2023

- Developed a 3D animal pose-reconstruction pipeline for biomechanics studies, matching **manual-level annotation accuracy** with 4× higher throughput on lizard locomotion datasets.
- Built a UV-based self-supervised data collection system (**110 samples/hour**) and applied U-Net for bag keypoint localization.
- Co-developed SLIP-Bagging, combining interactive perception and SlowFast network to achieve **67–81%** success rates in opening and bagging deformable bags across multiple material types.
- Contributed to the AVSI medical robotics framework, achieving **80–95%** success in vascular shunt insertion and **70–100%** success in vascular dilation, at **sub-millimeter accuracy** through visual servoing.

PROFESSIONAL SERVICE

Conference Reviewer for IEEE International Conference on Robotics and Automation (ICRA 2023, 2026); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2023).

SKILLS

- **Programming Languages:** Python, C/C++, MATLAB, Verilog, Java; familiar with Assembly, Bash, and SQL.
- **Machine Learning and Robotics: Frameworks:** PyTorch, TensorFlow, Keras; **Platforms:** ROS2, Gazebo, MuJoCo, PyBullet; **Methods:** CNNs, LSTMs, Transformers, imitation learning, reinforcement learning (PPO, residual RL).
- **Electronics Design and Simulation: PCB design:** Cadence Allegro, Altium Designer, KiCad; **Simulation:** LTspice, HSPICE.
- **Embedded Systems:** **MCUs:** ESP32, STM32, NXP (i.MX RT, MCX), Microchip PIC, Nordic nRF5x; **FPGA:** Xilinx Vivado; **Communication Protocol:** JTAG, SPI, I2C, UART, CAN, Ethernet.
- **Mechanical Design and Fabrication:** **CAD:** SolidWorks, AutoCAD, Onshape; **Fabrication:** 3D printing (FDM/SLA), CNC machining, laser cutting, soft lithography.
- **Experimental Methods and Characterization:** Instron, LCR & source meter, oscilloscope, network analyzer, DAQs, potentiostat.